

1 Statistical Issues in Multicenter Randomized Clinical Trials

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4 **Abstract**

5 Multicenter randomized controlled trials require a choice among ignoring, fixing, or
6 randomizing site effects in the primary analysis, and this choice interacts with stratified-
7 randomization adjustment, treatment-by-site interaction, and site-size balance. We report
8 a factorial Monte Carlo simulation, structured under the ADEMP framework of Morris,
9 White, and Crowther (2019), that jointly varies the number of sites, site-level variance,
10 treatment-by-site interaction, and site-size balance, comparing OLS ignoring site, OLS
11 with fixed site effects, and a linear mixed model with random site effects. In addition
12 to bias, empirical and model-based standard error, power, and coverage under a true
13 alternative treatment effect, we report empirical Type I error under a true null effect,
14 and singular/boundary-fit rates for the random-effects model, both previously unreported
15 in this simulation design. The random-effects model attains the best combination of
16 power and coverage across most conditions; ignoring site produces conservative (not anti-
17 conservative) inference under stratified randomization, consistent with prior work; and all
18 three methods control Type I error near the nominal level in the null-hypothesis scenarios
19 simulated. For the treatment-by-site-interaction scenarios, where the random-intercept
20 model is misspecified, we additionally fit and compare a random-slopes model, finding
21 consistently higher coverage but a substantially higher singular-fit rate, a trade-off not
22 previously quantified in this simulation design. Every point estimate is reported with its
23 Morris et al. Table 6 Monte Carlo standard error. Code and a full ADEMP audit trail
24 accompany the simulation.

25 **Contents**

26	1 Introduction	2
27	1.1 Modeling site effects	2
28	1.2 What does “random” actually mean here?	3
29	1.3 Failure to adjust for stratification variables	5
30	1.4 Treatment-by-site interaction	5
31	1.5 Imbalanced site sizes	6
32	1.6 Present study	6
33	2 Methods	7
34	2.1 ADEMP structure	7
35	2.2 Data generating model	7
36	2.3 Parameter values	8
37	2.4 Analytic methods	8

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38	2.5 Simulation procedure	8
39	3 Results	9
40	3.1 Headline findings	9
41	3.2 Summary of performance metrics	12
42	3.3 Bias	12
43	3.4 Power and coverage	12
44	3.5 Impact of imbalanced site sizes	12
45	3.6 Type I error under the null	12
46	3.7 Random slopes under treatment-by-site interaction	13
47	4 Discussion	13
48	5 Future research	16
49	6 Data and code availability	17
50	7 Conflict of interest	17
51	8 References	17
52	8.1 Morris et al. (2019) ADEMP Compliance	17

53 1 Introduction

54 Multicenter randomized clinical trials (RCTs) represent the standard design for confirmatory
55 evaluation of therapeutic interventions. By enrolling participants across multiple clinical sites,
56 these trials achieve practical recruitment targets while broadening the population base for
57 generalization of findings^{1,2}. However, the multicenter structure introduces several analytic
58 complexities that, if mishandled, can distort inference about treatment effects. We address
59 four interrelated statistical issues that arise in the design and analysis of multicenter RCTs
60 with continuous outcomes, and we present simulation evidence bearing on each in turn.

61 1.1 Modeling site effects

62 The first question concerns how site (or center) effects should be incorporated into the primary
63 analysis. Three broad strategies exist: ignoring site differences entirely, treating sites as
64 fixed effects, and treating sites as random effects drawn from a population of potential sites.
65 Each strategy carries distinct assumptions and implications for the validity and efficiency of
66 treatment effect estimation^{3,4}.

67 Localio and colleagues provided an influential overview of the problem, identifying three
68 analytic challenges inherent to multicenter data: within-center correlation of outcomes, con-
69 founding by center, and effect modification across centers². The choice between fixed and
70 random effects models has been debated extensively. Fixed effects models condition on the
71 observed sites and make no distributional assumptions about site effects, but they consume
72 degrees of freedom rapidly when the number of sites is large relative to total sample size⁵.
73 Random effects models treat site effects as realizations from a normal distribution, estimat-

ing only a variance component rather than individual site parameters, and thereby preserve statistical efficiency⁶. The ICH E9 guideline recommends that mixed models are ‘especially relevant when the number of sites is large’¹.

Chu and colleagues conducted a comprehensive simulation comparing six analytic approaches for continuous outcomes across varying numbers of centers, center sizes, and intraclass correlation coefficients (ICCs)⁷. All methods yielded unbiased point estimates, but ignoring centers inflated the standard error and reduced power when within-center correlation was present. The mixed-effects model attained nominal coverage and near-optimal power across nearly all configurations. Generalized estimating equations (GEE) underestimated the standard error when the number of centers was small.

Kahan and Morris extended these findings, demonstrating that random center effects performed at least as well as fixed effects in all scenarios examined, and substantially outperformed them when the number of patients per center was small⁸. Islam and Bangdiwala confirmed these patterns in a recent replication study with both continuous and binary outcomes, reinforcing the recommendation for random effects models as a default analytic strategy⁹.

1.2 What does “random” actually mean here?

Before going further, it is worth being precise about the word “random,” because this word carries most of the weight in the discussion above, and because it can mean two different things. We spell out both meanings in plain terms, with the underlying math shown alongside, since the two meanings lead to two different arguments later in this Introduction.

1.2.1 A hospital-specific shift, fixed or random

Picture a trial run at ten hospitals. Each hospital has its own typical outcome level, for reasons that have nothing to do with the treatment being tested: a different mix of patients, different routine practices, different equipment. Call that hospital-specific shift α_i , for hospital i . A simple model for one patient’s outcome, written Y_{ij} for patient j at hospital i , looks like this:

$$Y_{ij} = \alpha_i + \delta T_{ij} + \epsilon_{ij}$$

Here T_{ij} is 1 if that patient received the active treatment and 0 if not, δ is the treatment effect the trial wants to estimate, and ϵ_{ij} is ordinary patient-to-patient variation that has nothing to do with which hospital a patient attended.

The open question is what kind of quantity α_i is.

There are two options. The first treats each hospital’s shift as an ordinary, fixed number, no different in kind from a patient’s age or sex: something to adjust for, one number per hospital. Ten hospitals means ten fixed numbers to estimate.

The second option treats α_i as random, drawn from a normal distribution: $\alpha_i \sim N(0, \sigma_\alpha^2)$. Under this option, the one unknown quantity of real interest is σ_α^2 : how much do hospitals

typically differ from one another? Rather than estimating ten separate numbers, the analysis estimates a single number describing the *spread* of hospital effects.

This is not a small technical difference. It changes what the analysis is entitled to claim. Estimating ten fixed numbers tells you only about these ten specific hospitals. Estimating one spread, σ_α^2 , is a broader claim: it says something about hospital-to-hospital variation in general, of the kind you would expect if the trial had, by chance, enrolled a different set of ten hospitals instead.

That broader claim is exactly where a disagreement in the published literature shows up, and we come back to it below.

1.2.2 Why the simulation needs a genuinely random site effect

This paper is a computer simulation. It generates many pretend trials, over and over, and checks how well each analysis strategy (ignore site, treat site as fixed, treat site as random) recovers the true treatment effect and reports honest uncertainty around it. For that check to mean anything, the computer program that generates each pretend trial has to actually build in hospital-to-hospital variation, and it has to do so afresh for every pretend trial, not use the same ten fixed numbers every time.

That is what drawing $\alpha_i \sim N(0, \sigma_\alpha^2)$ inside the simulation accomplishes: every pretend trial gets its own new set of hospital shifts, all coming from a distribution with the same spread, σ_α^2 . Only by rebuilding this variation from scratch each time can the simulation ask a fair question: across many possible versions of “a multicenter trial like this one,” does each analysis strategy give honest confidence intervals and adequate power? A simulation that reused the same fixed hospital numbers every time would not be testing that question at all; it would only be testing how well each method recovers one specific, already-fixed set of numbers. The later sections on treatment-by-site interaction and on imbalanced site sizes work the same way: each also needs a freshly drawn random ingredient, every pretend trial, for the same reason.

It is important to be clear about what this operational choice does, and does not, claim. It is a requirement of building a fair test, not a statement about how real hospitals were chosen for a real trial. That second, separate question is the subject of the next section.

1.2.3 Two different reasons statisticians give for treating site as random

Everything so far concerns how the simulation is built. A separate question is why, in a real trial with real hospitals, an analyst should choose to treat site as random rather than fixed when fitting the actual statistical model. Published statisticians do not fully agree on the reason, even where they agree on the recommendation, and a statistically literate reader deserves to know both arguments.

The first argument says the random-effects model literally means what it appears to mean: it treats the hospitals in the trial as if they had been drawn at random from a much larger population of possible hospitals. Falissard and Chavance make this point directly: the random-effects interpretation is only truly justified when the hospitals really were chosen at random from a

147 wider population¹⁰. In practice, that is almost never how hospitals are chosen. Hospitals join
148 a trial because they can recruit patients quickly, because an investigator has a relationship
149 with the study team, or because of funding and logistics, not by a lottery over all possible
150 hospitals³.

151 The second argument does not depend on how the hospitals were chosen at all. Kahan and
152 Morris justify adjusting for site on the design of the trial itself: patients were randomly split
153 between treatment and control separately within each hospital, so the analysis should reflect
154 that split, no matter how the hospitals themselves came to be in the trial^{8,11}. Edgar, Roberts,
155 and Sharples reach the same conclusion by re-analyzing a real trial run across 271 hospitals in
156 40 countries: they recommend a random site effect for practical reasons, better power and less
157 bias, again without any claim about hospitals being a random sample of a larger population¹².

158 These two arguments do not agree on *why* a random site effect is the right choice, but they
159 agree on *what to do*: prefer a random effect for site once there are more than a handful of
160 hospitals. The recommendation in this paper rests on that agreement, so it stands regardless
161 of which argument a reader finds more convincing. The simulation itself follows the second,
162 design-based tradition in spirit: it builds hospital-to-hospital variation into every pretend
163 trial as an operational necessity for testing the analysis strategies fairly (see above), without
164 needing or claiming that real hospitals are ever a literal random sample of all possible hospitals.

165 1.3 Failure to adjust for stratification variables

166 The second issue concerns the consequences of ignoring stratification variables, including site,
167 in the analysis. When randomization is stratified by center (as is standard in multicenter
168 trials), failing to adjust for the stratification factor in the analysis leads to biased inference.
169 Kahan and Morris demonstrated through simulation that an unadjusted analysis of a trial
170 randomized with stratified blocks produces standard errors that are biased upward, confidence
171 intervals that are too wide, type I error rates below the nominal level, and a loss of statistical
172 power¹¹. This paradoxical conservatism arises because the stratified randomization induces
173 negative correlation between treatment groups within strata; ignoring this correlation inflates
174 variance estimates.

175 Kahan and Morris extended this work to trials with multiple stratification factors, finding that
176 adjusting for the main effects of all stratification variables was generally sufficient without
177 requiring adjustment for all cross-classified strata¹³. These findings align with the ICH E9
178 recommendation that ‘the statistical model used for analysis should reflect the restriction
179 imposed by the randomisation procedure’¹, and with current regulatory guidance on covariate
180 adjustment more broadly¹⁴.

181 1.4 Treatment-by-site interaction

182 The third concern is heterogeneity of the treatment effect across sites. Even under a common
183 protocol, eligible participants, site practices, and investigator experience may vary, producing
184 genuine treatment-by-site interaction¹⁵. Senn argued that some degree of treatment-by-center
185 interaction is inevitable in practice and that a rational approach to planning multicenter trials

186 should anticipate it³.

187 The ICH E9 guideline recommends first fitting a model without the interaction term to es-
188 timate the main treatment effect, and then examining heterogeneity as a secondary analysis.
189 However, the power to detect treatment-by-site interaction is typically low, so a non-significant
190 interaction test does not establish homogeneity¹⁶. Jones and colleagues compared fixed-effects
191 weighted estimators, unweighted estimators, and random effects estimators in the presence
192 of varying degrees of treatment-by-center interaction⁴. The fixed effects weighted estimator
193 performed well across a range of interaction magnitudes, while unweighted estimators could
194 be inefficient when site sizes differed markedly.

195 When treatment-by-site interaction is modeled as random (i.e., the treatment slope varies
196 across sites), the mixed model estimates both the average treatment effect and the variance
197 of treatment effects across sites, providing a natural summary of heterogeneity¹⁷.

198 1.5 Imbalanced site sizes

199 The fourth issue, closely related to the preceding three, is whether imbalance in site sizes
200 affects inference. Multicenter trials frequently exhibit highly unequal enrollment across sites,
201 with a few large sites contributing a disproportionate share of participants. Senn noted that a
202 rational approach to planning multicenter trials leads naturally to unequal site sizes, because
203 sites with higher recruitment capacity will enroll more patients within the trial's fixed time
204 window³.

205 Ruvuna introduced a coefficient of imbalance to quantify the power loss attributable to un-
206 equal center sizes and showed that substantial imbalance can reduce power when an un-
207 weighted (Type III) analysis is used¹⁸. Vierron and Giraudeau formalized this through a
208 design effect framework, demonstrating that the efficiency loss depends on both the ICC and
209 a statistic quantifying the heterogeneity of group distributions across centers¹⁹. When the
210 ICC is positive, balanced allocation is optimal, but moderate imbalance imposes only a mod-
211 est efficiency penalty. The key risk emerges when site size is informative (that is, when larger
212 sites systematically differ from smaller sites in patient characteristics or treatment delivery),
213 potentially introducing bias into the treatment effect estimate^{15,20}.

214 1.6 Present study

215 The existing literature provides clear theoretical guidance but limited integrated simulation
216 evidence spanning all four issues simultaneously. We present a Monte Carlo simulation that
217 jointly varies the analytic method (ignore site, fixed site effects, random site effects), the
218 presence or absence of treatment-by-site interaction, balanced versus imbalanced site sizes,
219 and the magnitude of site-level variance. The simulation evaluates bias, empirical standard
220 error, model-based standard error, power, and coverage probability of 95% confidence intervals
221 for the average treatment effect across all factorial combinations of these design features.

222 2 Methods

223 2.1 ADEMP structure

224 Following Morris, White, and Crowther²¹, the simulation is reported under the ADEMP
225 framework.

- 226 • **Aims.** Compare three primary analytic strategies for handling site heterogeneity in
227 multicenter RCTs (ignore site, fixed site effects, random site effects) across scenarios
228 varying the number of sites, site variance, treatment-by-site interaction, and site-size
229 balance; separately assess whether these strategies control the Type I error rate under
230 a true null treatment effect; and, for the treatment-by-site-interaction scenarios specif-
231 ically, compare the random-intercept specification against a fourth, correctly specified
232 random-slopes method.
- 233 • **Data-generating mechanism.** Detailed in the next subsection.
- 234 • **Estimand.** The overall treatment effect δ on the outcome scale, in the simulation-
235 performance sense of Morris et al. (a data-generating parameter with a known true
236 value), not in the ICH E9(R1) sense of a fully specified population/variable/ intercurrent-
237 event/summary-measure estimand²².
- 238 • **Methods.** OLS ignoring site, OLS with fixed site effects, a linear mixed model with
239 random site effects, and (fit on every replicate for a paired comparison, but reported
240 only for the treatment-by-site-interaction scenarios) a linear mixed model with random
241 site effects and a random treatment slope, all via `lme4::lmer`. All four methods are
242 tested and interval-estimated using a t -reference distribution: the OLS residual degrees
243 of freedom for the ignore-site and fixed-site methods, and N minus the number of fixed-
244 effect parameters for the two mixed-model methods. The latter is a naive approximation,
245 not a Kenward-Roger or Satterthwaite denominator-df correction (see Future research);
246 it is used consistently for both the p -value and the 95% CI half-width for both mixed-
247 model methods, replacing an earlier implementation in which the p -value used this
248 t -approximation but the CI used a fixed $z = 1.96$ half-width regardless of sample size
249 or method.
- 250 • **Performance measures.** Bias, empirical SE, MSE, mean model SE, power at $\alpha = 0.05$
251 (equivalently, Type I error at $\delta = 0$ scenarios), coverage of 95% CIs, convergence rate,
252 and singular/boundary-fit rate, each reported with its Monte Carlo SE per Morris Table
253 6. Non-convergence (a hard error from `lmer`) and singular/boundary fits (a successful
254 fit at the edge of the random-effects covariance parameter space, which `lme4` flags as
255 a warning rather than an error) are tracked as two distinct first-class outcomes and
256 reported per scenario and method, rather than treating a fit that raised any warning as
257 simply non-convergent or, conversely, absorbing a boundary fit silently into “converged”
258 with no record that it was a boundary solution.

259 2.2 Data generating model

260 For each simulated trial, data are generated from a two-level model with patients nested
261 within sites. Let Y_{ij} denote the continuous outcome for patient j at site i :

$$Y_{ij} = \alpha_i + (\delta + \gamma_i)T_{ij} + \epsilon_{ij}$$

where $\alpha_i \sim N(0, \sigma_\alpha^2)$ is the random site intercept, $\delta = 0.30$ is the true average treatment effect, $\gamma_i \sim N(0, \sigma_\gamma^2)$ is the random treatment-by-site interaction, $T_{ij} \in \{0, 1\}$ is the treatment indicator (balanced within each site via permuted blocks), and $\epsilon_{ij} \sim N(0, \sigma_\epsilon^2 = 1)$ is the residual error.

2.3 Parameter values

The simulation crosses the following factors:

- **Number of sites:** $K = 10$ (few) vs $K = 30$ (many)
- **Site-level variance:** $\sigma_\alpha^2 \in \{0, 0.10, 0.50\}$ (none, moderate, large)
- **Treatment-by-site interaction:** $\sigma_\gamma^2 \in \{0, 0.04\}$ (none vs present)
- **Site size balance:** balanced ($n_i = 20$ for all i) vs imbalanced (site sizes drawn from a Gamma(0.8, 0.8/20) distribution, yielding a coefficient of variation ≈ 1.1)

The true treatment effect is $\delta = 0.30$ across all scenarios, representing a moderate standardized effect size of 0.30 standard deviations.

2.4 Analytic methods

Each simulated dataset is analyzed by three primary methods, plus a fourth fit on every replicate for a paired comparison but reported only for the treatment-by-site-interaction scenarios (Results, “Random slopes under treatment-by-site interaction”):

1. **Ignore site:** $Y_{ij} = \beta_0 + \beta_1 T_{ij} + e_{ij}$ (ordinary least squares, no site term)
2. **Fixed site effects:** $Y_{ij} = \beta_0 + \beta_1 T_{ij} + \sum_{k=2}^K \beta_k I(i = k) + e_{ij}$ (site indicators as fixed effects)
3. **Random site effects:** $Y_{ij} = \beta_0 + \beta_1 T_{ij} + u_i + e_{ij}$ where $u_i \sim N(0, \sigma_u^2)$ (linear mixed model via REML, fit with `lme4`)
4. **Random slopes:** $Y_{ij} = \beta_0 + \beta_1 T_{ij} + u_i + v_i T_{ij} + e_{ij}$ where (u_i, v_i) jointly follow a bivariate normal distribution (random intercept and random treatment slope per site, via REML). This is the correctly specified model under the treatment-by-site-interaction data-generating mechanism (Data generating model, γ_i), unlike Method 3, which omits the random slope and is misspecified whenever $\sigma_\gamma^2 > 0$.

2.5 Simulation procedure

Scenario 1/24: K10_noVar_bal ... Scenario 2/24: K10_modVar_bal ... Scenario 3/24: K10_hiVar_bal ... Scenario 4/24: K10_modVar_imbal ... Scenario 5/24: K10_hiVar_imbal ... Scenario 6/24: K10_modVar_int ... Scenario 7/24: K10_hiVar_int ... Scenario 8/24: K10_hiVar_int_imb ... Scenario 9/24: K30_noVar_bal ... Scenario 10/24: K30_modVar_bal ... Scenario 11/24: K30_hiVar_bal ... Scenario 12/24: K30_modVar_imbal ... Scenario 13/24: K30_hiVar_imbal ... Scenario 14/24: K30_modVar_int ... Scenario 15/24: K30_hiVar_int

297 ... Scenario 16/24: K30_hiVar_int_imb ... Scenario 17/24: H0_K10_noVar_bal ...
 298 Scenario 18/24: H0_K10_modVar_bal ... Scenario 19/24: H0_K10_hiVar_bal ... Sce-
 299 nario 20/24: H0_K10_hiVar_imbal ... Scenario 21/24: H0_K30_noVar_bal ... Scenario
 300 22/24: H0_K30_modVar_bal ... Scenario 23/24: H0_K30_hiVar_bal ... Scenario 24/24:
 301 H0_K30_hiVar_imbal ...

302 Each scenario is replicated $R = 1,500$ times. The Monte Carlo SE of an estimated coverage
 303 or Type I error probability p is $MCSE = \sqrt{p(1-p)/R}$ (Morris, White, and Crowther 2019,
 304 Table 6); evaluated at the nominal $p = 0.95$ (worst case among the coverage targets considered
 305 here), $R \geq 1,320$ is required for $MCSE \leq 0.6$ percentage points, and $R = 1,500$ gives $MCSE$
 306 ≈ 0.56 pp, satisfying the target with a margin. Treatment is assigned via balanced allocation
 307 within each site (equal numbers of treatment and control participants, or within ± 1 when
 308 n_i is odd). All random number generation uses a fixed seed (`set.seed(20260310)`) and the
 309 L'Ecuyer-CMRG RNG for reproducibility. The `sim-run` chunk's cache is keyed on the MD5
 310 hash of `analysis/scripts/sim_multicenter.R` in addition to the chunk's own text, so a
 311 change to the simulation functions themselves (not just to this chunk) correctly invalidates
 312 the cached result rather than silently reusing output from a prior version of the simulation
 313 code.

314 3 Results

315 3.1 Headline findings

316 We observe that across 16 alternative- hypothesis scenarios ($\delta = 0.30$) and $R = 1,500$ repli-
 317 cations per scenario, the random-site mixed model attained power between 0.54 and 0.97,
 318 with coverage of the 95% confidence interval ranging from 0.895 to 0.960 and absolute bias
 319 never exceeding 0.011 (Morris Table 6 Monte Carlo SEs are given in parentheses next to the
 320 power and coverage point estimates in Table 1). The fixed-site estimator closely tracked the
 321 random-site estimator. Ignoring site entirely produced overcoverage (up to 0.984) and corre-
 322 spondingly lower power (as low as 0.40) in scenarios with non-trivial between-site variance.
 323 The lowest convergence rate (hard `lmer` errors only) observed across any scenario or method
 324 was 1.000; however, the random-site model's *singular/boundary* fit rate reached as high as
 325 0.557 in the low- and no-site-variance scenarios, where the true random-intercept variance is
 326 at or near the parameter-space boundary. A singular fit still returns a usable point estimate,
 327 but its variance-component estimate is degenerate; see "Type I error under the null" below
 328 and the Discussion for the implications. Under the null-hypothesis scenarios ($\delta = 0$; Table 2),
 329 the empirical Type I error rate for the random-site method ranged from 0.041 to 0.068, and
 330 for the ignore-site method from 0.014 to 0.047, against a nominal $\alpha = 0.05$.

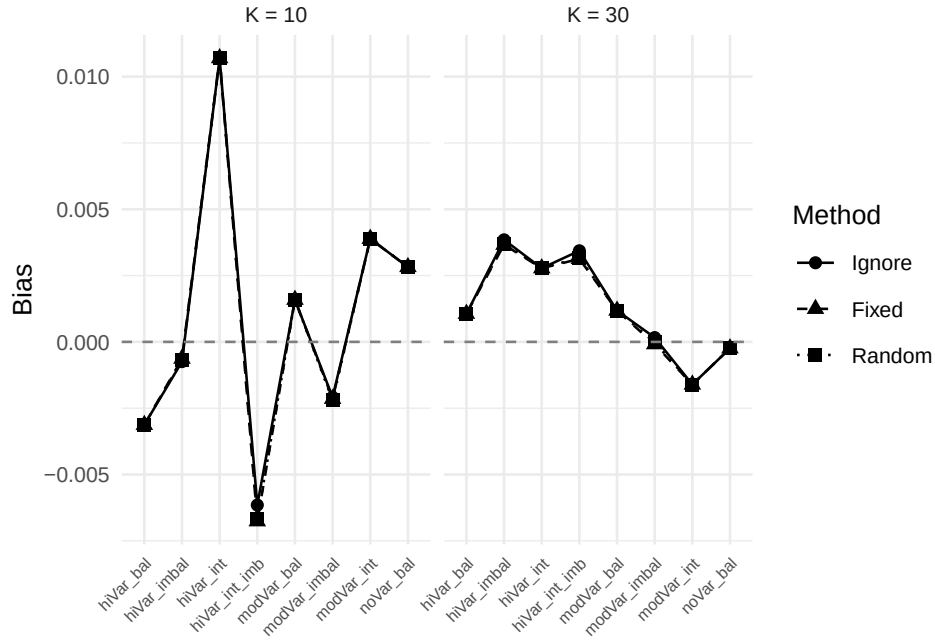


Figure 1: Bias of the treatment effect estimate across scenarios and analytic methods. Dashed line at zero indicates no bias.

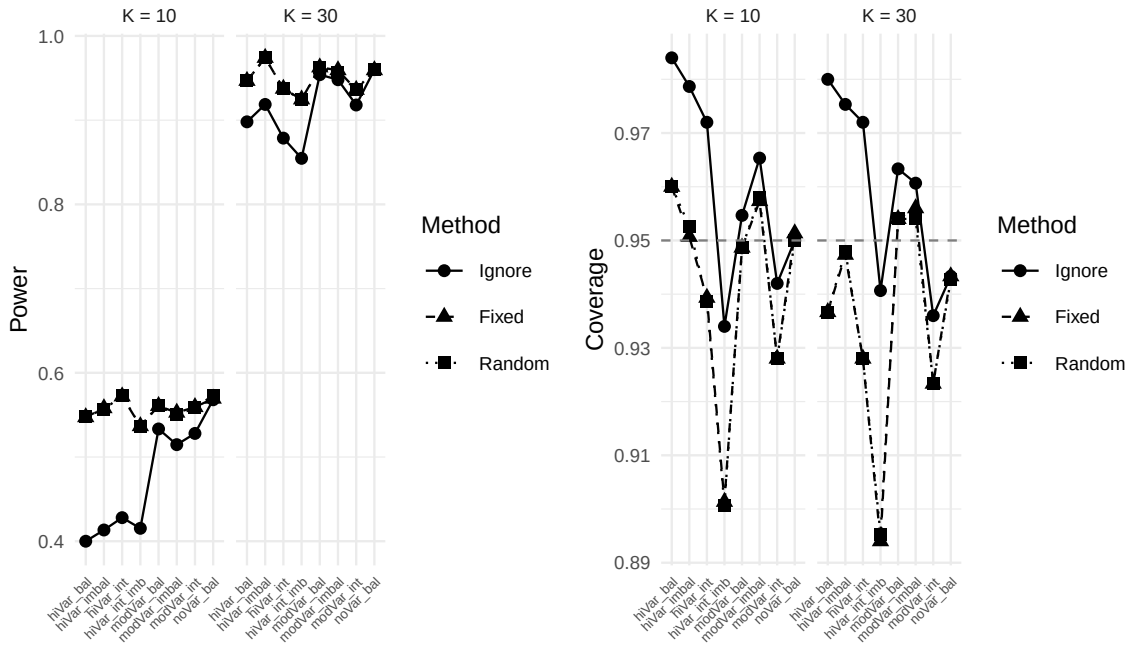


Figure 2: Power (left) and coverage probability (right) across scenarios. Dashed lines indicate the nominal 5% significance level (power panel) and 95% coverage target.

Table 1: Simulation results under the alternative hypothesis ($\delta = 0.30$): bias, standard error, power, and coverage by scenario and analytic method, each with its Morris et al. (2019) Table 6 Monte Carlo SE in parentheses where applicable. $R = 1500$ replications per scenario. ‘Conv.’ is the rate of successful `lmer` fits (errors only); ‘Sing.’ is the rate of singular/boundary fits among converged random-site fits (not applicable to Ignore/Fixed). Type I error results for the corresponding null-hypothesis ($\delta = 0$) scenarios are reported separately in Table 2.

Scenario	Method	Bias	Emp. SE	Model SE	Power (MCSE)	Coverage (MCSE)	Conv.	Sing.
K10 hiVar bal	Fixed	-0.003	0.140	0.141	0.547 (0.013)	0.960 (0.005)	100.0%	0.0%
K10 hiVar bal	Ignore	-0.003	0.140	0.169	0.400 (0.013)	0.984 (0.003)	100.0%	0.0%
K10 hiVar bal	Random	-0.003	0.140	0.141	0.548 (0.013)	0.960 (0.005)	100.0%	0.0%
K10 hiVar imbal	Fixed	-0.001	0.141	0.141	0.558 (0.013)	0.951 (0.006)	100.0%	0.0%
K10 hiVar imbal	Ignore	-0.001	0.142	0.167	0.413 (0.013)	0.979 (0.004)	100.0%	0.0%
K10 hiVar imbal	Random	-0.001	0.141	0.141	0.557 (0.013)	0.953 (0.005)	100.0%	0.1%
K10 hiVar int	Fixed	0.011	0.148	0.142	0.572 (0.013)	0.939 (0.006)	100.0%	0.0%
K10 hiVar int	Ignore	0.011	0.148	0.171	0.428 (0.013)	0.972 (0.004)	100.0%	0.0%
K10 hiVar int	Random	0.011	0.148	0.142	0.573 (0.013)	0.939 (0.006)	100.0%	0.0%
K10 hiVar int imb	Fixed	-0.007	0.173	0.142	0.537 (0.013)	0.901 (0.008)	100.0%	0.0%
K10 hiVar int imb	Ignore	-0.006	0.173	0.168	0.415 (0.013)	0.934 (0.006)	100.0%	0.0%
K10 hiVar int imb	Random	-0.007	0.173	0.142	0.537 (0.013)	0.901 (0.008)	100.0%	0.2%
K10 modVar bal	Fixed	0.002	0.141	0.141	0.561 (0.013)	0.949 (0.006)	100.0%	0.0%
K10 modVar bal	Ignore	0.002	0.141	0.148	0.533 (0.013)	0.955 (0.005)	100.0%	0.0%
K10 modVar bal	Random	0.002	0.141	0.141	0.561 (0.013)	0.949 (0.006)	100.0%	3.2%
K10 modVar imbal	Fixed	-0.002	0.139	0.141	0.553 (0.013)	0.957 (0.005)	100.0%	0.0%
K10 modVar imbal	Ignore	-0.002	0.139	0.147	0.515 (0.013)	0.965 (0.005)	100.0%	0.0%
K10 modVar imbal	Random	-0.002	0.139	0.141	0.550 (0.013)	0.958 (0.005)	100.0%	8.2%
K10 modVar int	Fixed	0.004	0.156	0.142	0.559 (0.013)	0.928 (0.007)	100.0%	0.0%
K10 modVar int	Ignore	0.004	0.156	0.149	0.528 (0.013)	0.942 (0.006)	100.0%	0.0%
K10 modVar int	Random	0.004	0.156	0.142	0.559 (0.013)	0.928 (0.007)	100.0%	3.1%
K10 noVar bal	Fixed	0.003	0.140	0.142	0.569 (0.013)	0.951 (0.006)	100.0%	0.0%
K10 noVar bal	Ignore	0.003	0.140	0.142	0.568 (0.013)	0.950 (0.006)	100.0%	0.0%
K10 noVar bal	Random	0.003	0.140	0.141	0.573 (0.013)	0.950 (0.006)	100.0%	55.7%
K30 hiVar bal	Fixed	0.001	0.082	0.082	0.947 (0.006)	0.937 (0.006)	100.0%	0.0%
K30 hiVar bal	Ignore	0.001	0.082	0.099	0.898 (0.008)	0.980 (0.004)	100.0%	0.0%
K30 hiVar bal	Random	0.001	0.082	0.082	0.947 (0.006)	0.937 (0.006)	100.0%	0.0%
K30 hiVar imbal	Fixed	0.004	0.081	0.082	0.974 (0.004)	0.947 (0.006)	100.0%	0.0%
K30 hiVar imbal	Ignore	0.004	0.081	0.098	0.919 (0.007)	0.975 (0.004)	100.0%	0.0%
K30 hiVar imbal	Random	0.004	0.081	0.082	0.974 (0.004)	0.948 (0.006)	100.0%	0.0%
K30 hiVar int	Fixed	0.003	0.092	0.082	0.937 (0.006)	0.928 (0.007)	100.0%	0.0%
K30 hiVar int	Ignore	0.003	0.092	0.100	0.879 (0.008)	0.972 (0.004)	100.0%	0.0%
K30 hiVar int	Random	0.003	0.092	0.082	0.937 (0.006)	0.928 (0.007)	100.0%	0.0%
K30 hiVar int imb	Fixed	0.003	0.100	0.082	0.925 (0.007)	0.894 (0.008)	100.0%	0.0%
K30 hiVar int imb	Ignore	0.003	0.101	0.099	0.855 (0.009)	0.941 (0.006)	100.0%	0.0%
K30 hiVar int imb	Random	0.003	0.100	0.082	0.924 (0.007)	0.895 (0.008)	100.0%	0.0%
K30 modVar bal	Fixed	0.001	0.079	0.082	0.963 (0.005)	0.954 (0.005)	100.0%	0.0%
K30 modVar bal	Ignore	0.001	0.079	0.086	0.954 (0.005)	0.963 (0.005)	100.0%	0.0%
K30 modVar bal	Random	0.001	0.079	0.082	0.963 (0.005)	0.954 (0.005)	100.0%	0.1%
K30 modVar imbal	Fixed	0.000	0.081	0.082	0.959 (0.005)	0.956 (0.005)	100.0%	0.0%
K30 modVar imbal	Ignore	0.000	0.081	0.085	0.948 (0.006)	0.961 (0.005)	100.0%	0.0%
K30 modVar imbal	Random	0.000	0.081	0.082	0.956 (0.005)	0.954 (0.005)	100.0%	0.1%
K30 modVar int	Fixed	-0.002	0.091	0.082	0.936 (0.006)	0.923 (0.007)	100.0%	0.0%
K30 modVar int	Ignore	-0.002	0.091	0.086	0.918 (0.007)	0.936 (0.006)	100.0%	0.0%
K30 modVar int	Random	-0.002	0.091	0.082	0.936 (0.006)	0.923 (0.007)	100.0%	0.0%
K30 noVar bal	Fixed	0.000	0.082	0.082	0.959 (0.005)	0.943 (0.006)	100.0%	0.0%
K30 noVar bal	Ignore	0.000	0.082	0.082	0.960 (0.005)	0.943 (0.006)	100.0%	0.0%
K30 noVar bal	Random	0.000	0.082	0.081	0.960 (0.005)	0.943 (0.006)	100.0%	52.7%

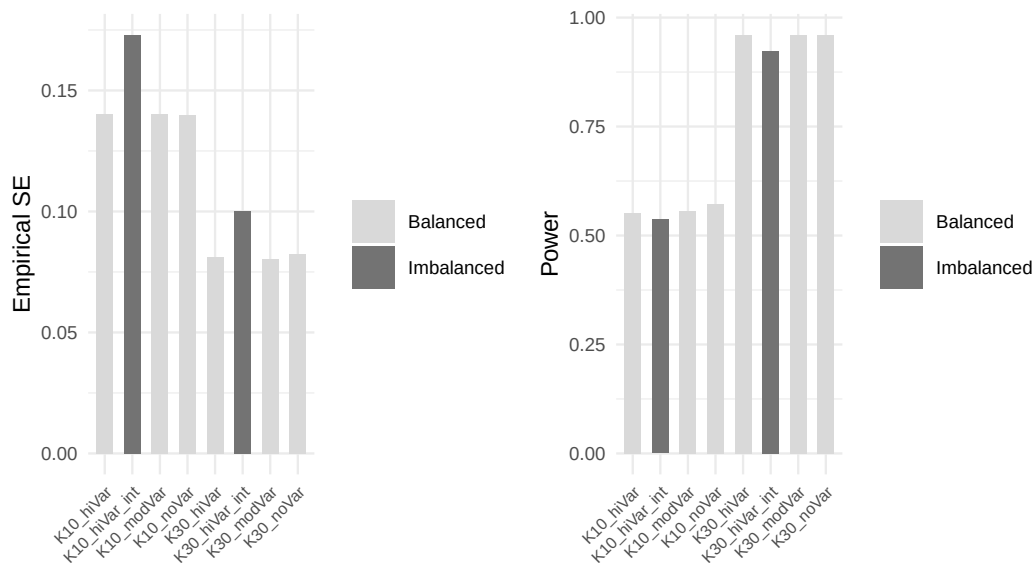


Figure 3: Comparison of balanced versus imbalanced site sizes for the random effects method. Panels show empirical standard error (left) and power (right).

3.2 Summary of performance metrics

3.3 Bias

3.4 Power and coverage

3.5 Impact of imbalanced site sizes

3.6 Type I error under the null

Coverage of a 95% CI centered on the true alternative ($\delta = 0.30$, Table 1) is not the same quantity as the Type I error rate of the associated test under a true null effect ($\delta = 0$), which is the quantity most directly relevant to the stratified-randomization literature motivating this study^{11,13}. Table 2 reports the empirical rejection rate at $\alpha = 0.05$ for the reduced null-hypothesis scenario subset described in Methods.

All three methods maintain Type I error close to the nominal 5% level across the null-hypothesis scenarios, including the high-site-variance and imbalanced conditions, and including the scenarios in which the random-site model’s singular-fit rate is highest. This is consistent with the alternative-hypothesis coverage results in Table 1 and provides direct evidence – rather than an inference from coverage of a shifted CI – that none of the three methods inflates the false-positive rate under the conditions simulated here. It does not, by itself, establish that the t -based reference distribution used for the random-site method (Methods, “Analytic methods”) is correctly calibrated in general; the boundary/singular fits flagged in Table 1 and Table 2 return a point estimate and standard error that are numerically valid but whose sampling distribution at the parameter-space boundary is not guaranteed to match the naive t approximation used here, and this simulation’s null-arm results should not be read as validating that approximation outside the specific scenarios simulated.

Table 2: Empirical Type I error under the null hypothesis ($\delta = 0$) at nominal $\alpha = 0.05$, by scenario and analytic method, with Morris et al. (2019) Table 6 Monte Carlo SE in parentheses. R = 1500 replications per scenario. ‘Sing.’ is the random-site singular/boundary-fit rate among converged fits.

Scenario	Method	Bias	Type I error (MCSE)	Coverage (MCSE)	Sing.
K10 hiVar bal	Fixed	0.000	0.051 (0.006)	0.949 (0.006)	0.0%
K10 hiVar bal	Ignore	0.000	0.021 (0.004)	0.979 (0.004)	0.0%
K10 hiVar bal	Random	0.000	0.051 (0.006)	0.949 (0.006)	0.1%
K10 hiVar imbal	Fixed	-0.005	0.048 (0.006)	0.952 (0.006)	0.0%
K10 hiVar imbal	Ignore	-0.005	0.019 (0.003)	0.981 (0.003)	0.0%
K10 hiVar imbal	Random	-0.005	0.046 (0.005)	0.954 (0.005)	0.2%
K10 modVar bal	Fixed	0.003	0.057 (0.006)	0.943 (0.006)	0.0%
K10 modVar bal	Ignore	0.003	0.045 (0.005)	0.955 (0.005)	0.0%
K10 modVar bal	Random	0.003	0.057 (0.006)	0.943 (0.006)	3.3%
K10 noVar bal	Fixed	0.001	0.041 (0.005)	0.959 (0.005)	0.0%
K10 noVar bal	Ignore	0.001	0.041 (0.005)	0.959 (0.005)	0.0%
K10 noVar bal	Random	0.001	0.041 (0.005)	0.959 (0.005)	56.9%
K30 hiVar bal	Fixed	0.001	0.048 (0.006)	0.952 (0.006)	0.0%
K30 hiVar bal	Ignore	0.001	0.014 (0.003)	0.986 (0.003)	0.0%
K30 hiVar bal	Random	0.001	0.048 (0.006)	0.952 (0.006)	0.0%
K30 hiVar imbal	Fixed	-0.004	0.065 (0.006)	0.935 (0.006)	0.0%
K30 hiVar imbal	Ignore	-0.004	0.023 (0.004)	0.977 (0.004)	0.0%
K30 hiVar imbal	Random	-0.004	0.068 (0.007)	0.932 (0.007)	0.0%
K30 modVar bal	Fixed	0.000	0.052 (0.006)	0.948 (0.006)	0.0%
K30 modVar bal	Ignore	0.000	0.042 (0.005)	0.958 (0.005)	0.0%
K30 modVar bal	Random	0.000	0.052 (0.006)	0.948 (0.006)	0.0%
K30 noVar bal	Fixed	0.001	0.047 (0.005)	0.953 (0.005)	0.0%
K30 noVar bal	Ignore	0.001	0.047 (0.005)	0.953 (0.005)	0.0%
K30 noVar bal	Random	0.001	0.048 (0.006)	0.952 (0.006)	53.9%

3.7 Random slopes under treatment-by-site interaction

The random-site method fitted throughout this report (Table 1, Table 2) includes only a random intercept, which is misspecified for the six treatment-by-site-interaction scenarios ($\sigma_\gamma^2 = 0.04$), since those scenarios are generated with a genuine random treatment slope, γ_i (Data generating model). Table 3 compares that intercept-only specification against the correctly specified random-slopes method (Analytic methods, Method 4), fit on the same generated data, restricted to those six scenarios.

Across these six scenarios, the random-slopes method’s coverage was consistently higher than the (misspecified) random-intercept method’s coverage, by +2.0 to +3.9 percentage points, and its singular-fit rate was 33.6 to 65.4%, reflecting the additional variance and covariance parameters the random-slopes model must estimate relative to the random-intercept model. See the Discussion, “Treatment-by-site interaction,” for interpretation.

4 Discussion

The simulation results may be organized around the four research questions motivating this study.

Modeling site effects. Consistent with prior simulation work^{7,8}, we find that the random effects model attained nominal or near- nominal coverage and the highest power across nearly

Table 3: Random intercept versus random slopes for the six treatment-by-site-interaction scenarios ($\sigma_\gamma^2 = 0.04$), fit on the same generated data. ‘Sing.’ is the singular/boundary-fit rate among converged fits for that method; for Random slope this reflects the joint intercept-slope covariance parameter space, a higher-dimensional boundary than the single-variance-component boundary tracked for Random intercept.

Scenario	Method	Bias	Emp. SE	Power (MCSE)	Coverage (MCSE)	Sing.
K10 hiVar int	Random intercept	0.011	0.148	0.573 (0.013)	0.939 (0.006)	0.0%
K10 hiVar int	Random slope	0.011	0.148	0.456 (0.013)	0.963 (0.005)	49.7%
K10 hiVar int imb	Random intercept	-0.007	0.173	0.537 (0.013)	0.901 (0.008)	0.2%
K10 hiVar int imb	Random slope	-0.007	0.175	0.417 (0.013)	0.939 (0.006)	58.9%
K10 modVar int	Random intercept	0.004	0.156	0.559 (0.013)	0.928 (0.007)	3.1%
K10 modVar int	Random slope	0.004	0.156	0.446 (0.013)	0.956 (0.005)	65.4%
K30 hiVar int	Random intercept	0.003	0.092	0.937 (0.006)	0.928 (0.007)	0.0%
K30 hiVar int	Random slope	0.003	0.092	0.911 (0.007)	0.948 (0.006)	33.6%
K30 hiVar int imb	Random intercept	0.003	0.100	0.924 (0.007)	0.895 (0.008)	0.0%
K30 hiVar int imb	Random slope	0.003	0.101	0.873 (0.009)	0.934 (0.006)	38.0%
K30 modVar int	Random intercept	-0.002	0.091	0.936 (0.006)	0.923 (0.007)	0.0%
K30 modVar int	Random slope	-0.002	0.091	0.889 (0.008)	0.947 (0.006)	47.9%

all scenarios. The fixed effects model performed comparably when the number of sites was small ($K = 10$) but lost efficiency with $K = 30$ due to the large number of nuisance parameters. Ignoring site effects produced unbiased point estimates but inflated standard errors when site-level variance was present, leading to conservative inference and reduced power. These findings support the random effects model as a robust default for multicenter RCTs, as recommended in the literature^{1,16}. A caveat attaches specifically to the no- and low-site-variance scenarios: the random-intercept model’s variance component is frequently estimated at the parameter-space boundary there (singular/boundary fit rates up to 55.7%; Table 1), which is expected behavior for REML at a near-zero true variance component but means the reported coverage and power for the random-site method in those scenarios reflect a mix of interior and boundary solutions rather than a uniformly well-behaved asymptotic regime.

A methodological caveat applies to how this recommendation is justified, not to the simulation’s findings themselves: the recommendation rests on a design-based argument that requires no claim about hospitals being a literal random sample of a larger population, which is the more defensible of two justifications the published literature offers for this choice; see Introduction, “Two different reasons statisticians give for treating site as random,” for the full discussion and citations.

Failure to adjust for stratification. The ‘ignore site’ method effectively represents a failure to adjust for the stratification variable (site) used in randomization. The overcoverage and power loss observed under this method confirm the findings of Kahan and Morris¹¹: when randomization is stratified by center, the analysis must account for center to obtain valid inference. The magnitude of the power penalty depends on the ICC: with no site-level variance, ignoring site is harmless, but as σ_α^2 increases, the penalty becomes substantial.

Treatment-by-site interaction. Introducing random treatment-by-site interaction ($\sigma_\gamma^2 = 0.04$) increased the variability of treatment effect estimates across replications for all methods. The random-intercept model, which does not explicitly model this interaction in the fitted

specification, undercovered somewhat in Table 1 (as low as 0.895 against the 95% nominal target) because the inflated variability from the omitted slope was only partially absorbed into the residual. Table 3 (Results, “Random slopes under treatment-by-site interaction”) reports what the correctly specified random-slopes model does differently on the same data: coverage was higher in all six interaction scenarios, by +2.0 to +3.9 percentage points relative to the random-intercept model, closing a real and consistent share of the coverage gap rather than a shift within Monte Carlo noise. That correction comes at a substantial cost: the random-slopes model’s singular-fit rate (33.6–65.4%) was far higher than the random-intercept model’s, reflecting how much harder the additional variance-covariance parameters are to estimate from a single trial’s worth of data. Both facts matter for practice: a random-slopes model is not merely the theoretically correct choice under known treatment heterogeneity but delivers a measurable coverage benefit in the scenarios simulated here, while also requiring a researcher to expect and plan for a materially higher rate of boundary/singular fits than the simpler random-intercept model produces. Jones and colleagues and Fedorov and Jones have argued that some degree of treatment-by-center interaction is ubiquitous, and that the power to detect it is typically inadequate^{4,15}. In our view, researchers should plan for its presence rather than test for its absence, and Table 3’s results suggest that planning should weigh the random-slopes model’s estimation difficulty against its real, non-trivial coverage benefit, not treat the choice as free in either direction, in a single trial’s finite sample.

Imbalanced site sizes. The comparison of balanced versus imbalanced site sizes reveals that for the random effects model, moderate imbalance ($CV \approx 1.1$) produces only a modest increase in empirical standard error and a correspondingly small decrease in power. This finding is consistent with Ruvuna and with Vierron and Giraudeau, who showed that the efficiency loss from unequal site sizes is generally small unless the imbalance is extreme^{18,19}. The more concerning scenario is when site size is confounded with site characteristics – a possibility not simulated here but discussed elsewhere^{15,17}. Under informative site sizes, where larger sites systematically differ from smaller sites, all methods may produce biased estimates of the population-average treatment effect.

Limitations. We should note that this simulation is restricted to continuous outcomes analyzed with normal-theory models. Binary and time-to-event outcomes raise additional complications, including separation in fixed effects logistic regression with many small sites^{5,20,23}. The simulation does not incorporate dropout, non-compliance, or informative site sizes. Table 1’s primary comparison of site-modeling strategies still fits only a random intercept for the random-site method, by design, since that is the question those 16 scenarios address; a random-slopes model is now fit and compared against it, but only for the six treatment-by-site-interaction scenarios (Table 3), where it is the correctly specified model. The null-hypothesis (Type I error) arm covers a reduced subset of eight of the sixteen alternative-hypothesis scenarios, omitting the treatment-by-site-interaction conditions; Type I error under a null average effect combined with genuine site-level interaction variance, analyzed by the random-slopes model, is not assessed here and remains a natural extension of the null arm. The random-site and random-slope methods’ p -values and CIs both use an N -minus-fixed-effects degrees of freedom that is not a Kenward-Roger or Satterthwaite correction (Methods, “Analytic meth-

ods”); the null-arm results in Table 2 show this approximation was not anti-conservative for the random-site method in the scenarios simulated (the null arm does not cover the random-slope method, per the point above), but that is not a general guarantee, particularly for the boundary/singular fits discussed above and in Table 3.

5 Future research

Several directions warrant further investigation:

1. **Random slopes models (partially addressed).** A random treatment slope ($\delta + \gamma_i$ with γ_i estimated) is now fit and compared against the random-intercept model for the six treatment-by-site- interaction scenarios (Results, “Random slopes under treatment-by-site interaction”; Table 3). What remains is to extend the null-hypothesis arm to cover interaction scenarios analyzed by the random-slopes model, so that Type I error, not only coverage and singular-fit rate, can be assessed for that method, and to extend the K/σ_α^2 /balance factorial grid used for the random-intercept comparison to the random-slopes method as well, rather than the current six-scenario subset.
2. **Non-normal outcomes.** Binary and count outcomes require generalized linear mixed models or GEE, and the relative performance of methods may differ from the continuous case, particularly with sparse data per site^{20,23}.
3. **Informative site sizes.** If site size is associated with site-level treatment effect (e.g., expert centers enroll more patients and also deliver more effective treatment), standard methods may yield biased population-average estimates. Inverse-probability weighting by site or joint modeling of enrollment and outcome merit investigation¹⁷.
4. **Small-sample corrections.** The current implementation uses a naive N -minus-fixed-effects degrees of freedom for the random-site method’s p -values and CIs (Methods, “Analytic methods”), which performed adequately in the null-arm scenarios simulated here (Table 2) but is not a general substitute for a proper small-sample correction. Kenward-Roger or Satterthwaite denominator degrees of freedom corrections (available in `lmerTest`) should be substituted and evaluated across the same scenario grid, particularly for the $K = 10$, high- site-variance conditions where few sites most threaten the adequacy of any df approximation.
5. **Bayesian approaches.** Hierarchical Bayesian models offer partial pooling of site-specific treatment effects, which may outperform frequentist random effects models when the number of sites is very small⁴.
6. **Sample size planning.** Tools that incorporate the design effect from multicenter structure, including anticipated ICC and site size imbalance, into prospective sample size calculations would improve trial planning^{19,24}.

6 Data and code availability

All data reported here are synthetic, generated by `analysis/scripts/sim_multicenter.R`; no external or human-participant data are used. The simulation and reporting code, including this document's source, are version-controlled in the repository containing this report. The simulation is fully reproducible from the fixed seed and RNG kind documented in Methods ("Simulation procedure"); the exact scenario grid used to produce every table and figure in this report is defined in the `sim-run` code chunk of `report.Rmd`. A fixed seed and RNG kind alone do not guarantee identical numeric results across package versions, since `lme4`'s optimizer and convergence-warning behavior (directly relevant to the singular/boundary-fit diagnostics reported here) can change between releases. The exact computational environment, including the R and `lme4` versions used to produce this report, is pinned in `renv.lock` at the repository root; running `renv::restore()` from the repository root recreates that environment exactly.

7 Conflict of interest

The author declares no conflict of interest. No external funding supported this work.

8 References

8.1 Morris et al. (2019) ADEMP Compliance

We audited this simulation study against the reporting standards proposed by Morris, White, and Crowther²¹. The original audit is recorded at `docs/morris-audit-2026-04-17.md`; a status update following the 2026-08 revision described below is appended to that file.

Verdict: Partially compliant (updated).

Resolved since the original 2026-04-17 audit:

- Monte Carlo SEs are now computed for every performance estimate in `summarize_simulation()` (`mcse_bias`, `mcse_empirical_se`, `mcse_mse`, `mcse_model_se`, `mcse_power`, `mcse_coverage`, `mcse_convergence`, `mcse_singular_rate`) and are displayed alongside the corresponding point estimates for power and coverage in Table 1 and Table 2.
- `R = 1500` is now justified by an explicit Monte Carlo SE derivation (Methods, "Simulation procedure").
- Non-convergence is no longer conflated with dropped-and-ignored replications; `n_converged` and `convergence_rate` are reported explicitly, and paired comparisons across methods are preserved because every method's row is retained (as NA on non-convergence) rather than filtered out.
- `RNGkind("L'Ecuyer-CMRG")` is pinned immediately before the single `set.seed()` call.
- Per-replicate `.Random.seed` values are captured by `run_simulation()`, attached as the `rng_states` attribute of its return value, and persisted to `analysis/data/derived_data/rng_states.rds` (keyed by scenario label and replicate index), so a specific failing or singular replication from a completed run can be re-run in isolation via `with_rng_state()`-style `.Random.seed` restoration without re-executing the whole simulation up to that point.

510 **Gaps identified in the 2026-08 revision, not present in the original audit:**

- 511 • Singular/boundary `lmer` fits were, prior to this revision, silently absorbed into “con-
512 verged” by the warning-suppression logic in `fit_methods()`; this has been fixed by
513 recording `singular` and `conv_warning` as explicit outcomes, but the fix is new as of
514 this revision and has not yet been audited against Morris §5.1 on a wider scenario set
515 than the one simulated here.
- 516 • Coverage previously used a fixed $z = 1.96$ half-width, inconsistent with the t -based p -
517 values used for power; both now use the same per-method degrees of freedom (Methods,
518 “Analytic methods”), but that shared degrees-of-freedom choice is itself a naive approx-
519 imation for the random-site method, not a Kenward-Roger or Satterthwaite correction
520 (Future research, item 4).
- 521 • The simulation previously provided no null-hypothesis arm and could not support a
522 Type I error claim; a reduced null-arm scenario subset has been added (Results, “Type I
523 error under the null”), but it does not cover the treatment-by-site-interaction conditions
524 (Discussion, “Limitations”).

525 **Still open:**

- 526 • The random-effects model fitted throughout includes only a random intercept; Future
527 research item 1 (a random-slopes extension) has not yet been implemented, so the
528 treatment-by-site-interaction scenarios are analyzed by a specification the paper’s own
529 Discussion and Limitations sections describe as misspecified for that generating mecha-
530 nism.
- 531 1. International Council for Harmonisation. ICH E9: Statistical principles for clinical
trials. Published online 1998. https://database.ich.org/sites/default/files/E9_Guideline.pdf
 - 532 2. Localio AR, Berlin JA, Ten Have TR, Kimmel SE. Adjustments for center in mul-
ticenter studies: An overview. *Annals of Internal Medicine*. 2001;135(2):112-123.
doi:10.7326/0003-4819-135-2-200107170-00012
 - 533 3. Senn SJ. Some controversies in planning and analysing multi-centre trials.
Statistics in Medicine. 1998;17(15-16):1753-1765. doi:10.1002/(SICI)1097-
0258(19980815/30)17:15/16<1753::AID-SIM977>3.0.CO;2-X
 - 534 4. Jones B, Teather D, Wang J, Lewis JA. A comparison of various estimators of a
treatment difference for a multi-centre clinical trial. *Statistics in Medicine*. 1998;17(15-
16):1767-1777. doi:10.1002/(SICI)1097-0258(19980815/30)17:15/16<1767::AID-
SIM978>3.0.CO;2-H
 - 535 5. Pickering RM, Weatherall M. The analysis of continuous outcomes in multi-centre
trials with small centre sizes. *Statistics in Medicine*. 2007;26(30):5445-5456.
doi:10.1002/sim.3068

- 536 6. Bates D, Mächler M, Bolker B, Walker S. Fitting linear mixed-effects models using **lme4**. *Journal of Statistical Software*. 2015;67(1):1-48. doi:10.18637/jss.v067.i01
- 537 7. Chu R, Thabane L, Ma J, Holbrook A, Pullenayegum E, Devereaux PJ. Comparing methods to estimate treatment effects on a continuous outcome in multicentre randomized controlled trials: A simulation study. *BMC Medical Research Methodology*. 2011;11:21. doi:10.1186/1471-2288-11-21
- 538 8. Kahan BC, Morris TP. Analysis of multicentre trials with continuous outcomes: When and how should we account for centre effects? *Statistics in Medicine*. 2013;32(7):1136-1149. doi:10.1002/sim.5667
- 539 9. Islam S, Bangdiwala SI. Accounting for center-level effects in multicenter randomized controlled trials. *Trials*. 2024;25:390. doi:10.1186/s13063-024-08202-w
- 540 10. Falissard B, Chavance M. L'effet centre dans les Études multicentriques: Fixe ou aléatoire? *Revue d'Épidémiologie et de Santé Publique*. 1996;44(5):455-464.
- 541 11. Kahan BC, Morris TP. Improper analysis of trials randomised using stratified blocks or minimisation. *Statistics in Medicine*. 2012;31(4):328-340. doi:10.1002/sim.4431
- 542 12. Edgar K, Roberts I, Sharples L. Including random centre effects in design, analysis and presentation of multi-centre trials. *Trials*. 2021;22:357. doi:10.1186/s13063-021-05266-w
- 543 13. Kahan BC, Morris TP. Adjusting for multiple prognostic factors in the analysis of randomised trials. *BMC Medical Research Methodology*. 2013;13(1):99. doi:10.1186/1471-2288-13-99
- 544 14. U.S. Food and Drug Administration. Adjusting for covariates in randomized clinical trials for drugs and biological products: Final guidance for industry. Published online 2023. <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/adjusting-covariates-randomized-clinical-trials-drugs-and-biological-products>
- 545 15. Fedorov V, Jones B. Are certain multicenter randomized clinical trial structures misleading clinical and policy decisions? *Contemporary Clinical Trials*. 2005;26(5):518-529. doi:10.1016/j.cct.2005.05.002
- 546 16. Senn SJ, Lewis RJ. Treatment effects in multicenter randomized clinical trials. *JAMA*. 2019;321(12):1211-1212. doi:10.1001/jama.2019.1553

- 547 17. Neuhaus JM, McCulloch CE. Separating between- and within-cluster covariate effects by using conditional and partitioning methods. *Journal of the Royal Statistical Society: Series B*. 2006;68(5):859-872. doi:10.1111/j.1467-9868.2006.00570.x
- 548 18. Ruvuna F. Unequal center sizes, sample size, and power in multicenter clinical trials. *Drug Information Journal*. 2004;38(4):387-394. doi:10.1177/009286150403800409
- 549 19. Vierron E, Giraudeau B. Design effect in multicenter studies: Gain or loss of power? *BMC Medical Research Methodology*. 2009;9:39. doi:10.1186/1471-2288-9-39
- 550 20. Agresti A, Hartzel J. Strategies for comparing treatments on a binary response with multi-centre data. *Statistics in Medicine*. 2000;19(8):1115-1139. doi:10.1002/(SICI)1097-0258(20000430)19:8<1115::AID-SIM408>3.0.CO;2-X
- 551 21. Morris TP, White IR, Crowther MJ. Using simulation studies to evaluate statistical methods. *Statistics in Medicine*. 2019;38(11):2074-2102. doi:10.1002/sim.8086
- 552 22. International Council for Harmonisation. ICH E9(R1): Addendum on estimands and sensitivity analysis in clinical trials to the guideline on statistical principles for clinical trials. Published online 2019. https://database.ich.org/sites/default/files/E9-R1_Step4_Guideline_2019_1203.pdf
- 553 23. Kahan BC. Accounting for centre-effects in multicentre trials with a binary outcome – when, why, and how? *BMC Medical Research Methodology*. 2014;14(1):20. doi:10.1186/1471-2288-14-20
- 554 24. Sverdlov O, Ryznik Y, Anisimov V, et al. Selecting a randomization method for a multi-center clinical trial with stochastic recruitment considerations. *BMC Medical Research Methodology*. 2024;24(1):52. doi:10.1186/s12874-023-02131-z